

BISWAJEET SAHU

Data Engineer

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Data Engineer with 3 years of experience in building scalable Data Pipelines, cloud Datawarehouse solutions, ETL and ELT, Big Data Processing, PowerBI dashboards and managing analytical operations using complex automated SQL queries to identify and provide business solutions as REPORTs using BI tools .

SKILLS

- **Languages:** Java, Python, SQL, Unix Shell Scripts, Linux, Spark
- **AWS:** EC2, S3, Lambda, Glue, Redshift, Quicksight, Athena, Kinesis
- **AZURE:** Azure Databricks, Azure Data Factory, Azure Synapse
- **Big Data & ETL:** Databricks, HDFS, PySpark, Hive, YARN, SparkSQL, Airflow, CICD Pipelines, Snowflake, Snowpipe, Talend, Kafka
- **Project Management:** JIRA
- **Analytics:** Talend, AWS Athena, Data Build Tool (DBT), Cloudera Data Lake
- **Databases:** MSSQL, H2, MySQL, PostgreSQL, Graph DB like Oracle NoSQL DB
- **Visualization:** Excel, PowerBI
- **Deployment:** Docker, Kubernetes
- **Math:** Statistics, Calculus, Probability, PNC, Matlab, Linear Algebra

EXPERIENCE

Jan 2022- July 2023 (19 months)

DATA ENGINEER at **HCL TECHNOLOGIES**

- **ELT in Data Migration Project**
 - Data ingestion from **Azure blob storage** to **Azure Data Lake Gen2** using Copy Activity
 - Automated workflow scheduled using **Event Trigger** for running **ADF data pipelines** once the file gets uploaded into Azure Blob storage
 - Performing transformations using **Azure Data Flow** once the data gets ingested into Azure Data Lake
 - Performing complex transformations using **Azure Databricks** notebooks which is coded with both Spark and SQL using **magic commands**. Also **databricks utilities** are used very frequently whenever required.
 - Using **Azure vaults backed by Secure scopes** where secrets are kept secured for accessing files inside containers of Data Lake storage using **mounting**
 - Using DBFS root for analyzing small files and keeping more frequently used data

- Using Access Keys,SAS token,Service Principal,Cluster scoped and AAD credentials passthrough authentication using Azure vaults backed with Secret scopes whenever suitable in notebooks.
- Automating all transformation steps in many notebooks using spark and calling dependent notebooks with a single notebook and scheduling it to run on particular interval.
- Using MSSQL DB for storing transformed results in the form of tables with predefined schemas.
- Visualization of data using POWERBI
- Sharing reports and interactive dashboards created using POWERBI on ADF
- Worked on Azure API Management platform for designing REST APIs
- Collaborating with Data Analyst and Data Scientists for data related use cases
- Experience developing Azure Data Factory (covering Azure Functions, LogicApps, Triggers, IR) , Databricks (PySpark), Stream Analytics, Event Hub & HD Insight Components
- Experience managing source data access security, using Vault, configuring authentication and authorization, enforcing data policies and standards
- **Data Analysis**
 - Automated the process of **ingesting, transforming, and loading data** into an AWS S3 bucket using **EMR clusters and PySpark scripting**. The entire workflow is **orchestrated** and scheduled with **Airflow**.
 - Ingestion using **batch data pipeline** from shared **external sources**(shared drive) to **AWS S3** bucket using **shell script** run by EMR clusters.
 - Automated flat file loaded from **AWS S3** to **Snowflake** table with the help of **snowpipe** and Notification services and loading them into staging layer.
 - Carrying End to end data transformations pipelines using **DBT**.
 - Writing **python** scripts for **ingestion and Loading data** managed using Airflow DAGs.
 - Proficiency in **Java scripting** for Transformation of data within ETL tools – **Talend** and **KNIME**.
 - Use of **Spark** clusters for managing **Big data (Transformations and Loading)**.**Databricks** platform is used for running Spark scripts for Big Data management. Use of **Azure Data Factory** services for managing end to end ETL data flow.
 - Proficiency in **Snowflake Data Cloud, Google BigQuery, DataBricks Lakehouse, Azure Synapse** for creating cloud Data warehouse.
 - Hands on knowledge in **MySQL,MSSQL,Oracle and PostgreSQL** for running complex SQL queries while data analysis.
 - Automation of **Transformation process using tasks**(performed using SQL queries) which runs once file gets ingested in staging layer and create schema and tables automatically inside Consumption(Output) layer.
 - Designed **PowerBI reports** using results performed on integration of PowerBI with Snowflake for Visualization.

- Tracking Log files generated using **RegEx** used in Python scripts.
- Exporting the resulting schema and tables back to the shared drive using Automated tasks ran on stored procedures using **SQL queries**.
- Experience in Container Management systems like **Kubernetes**.
- **Security of data**
 - Access management by creating custom users and roles
 - Grant usage and selected roles to those users
 - Manage security protocols like **SFTP** protocols while data transfers
- **Data Transformation and Visualization**
 - Designed and built a data Transformation process with 19 workflows on **KNIME** automated using batchscripts.
 - **Visualization** of transformed and cleaned data using **PowerBI(Generated 150+ reports on different datasets of varying types)**.
 - Running **100+ SQL queries on daily basis** for understanding huge datasets.
 - **Lead and mentored a member** for bug fixing and designing new pipelines as required.
 - Experience with **AWS,AZURE** and **GCP** services(mainly AWS and AZURE).
 - Cleaning and Transforming huge data using **DBT** for **data Analysis**,deploying changes in code to GitHub using pipelines after testing in **DBT cloud** and Maintain **data documentation** and definitions within **dbt** as they build and develop lineage graphs.

Oct 2020-Dec 2021 (14 months)

Data Analyst at FSNL

- Designing data pipelines using PySpark .
- Data analysis in databricks Notebooks using Python libraries like pandas,numpy,etc
- Manage and optimize data in **Google BigQuery** to ensure accuracy, performance, and security.
- Develop strategies for **ingesting data** from files and other structured or unstructured sources.
- Work with **Cube Flow** to create data cubes and enhance data querying performance.
- Design and schedule data workflows using **Apache Airflow** for automation and orchestration.
- Experience with Cloudera Data Lake

EDUCATIONS:

- **2014 – 2018**
BTECH , Mizoram University, Electrical Engineering - 8.73 GPA
- **2011-2013**
High School from Kendriya Vidyalaya Surda - 87.80%
- **2010-2011**
Intermediate from St. Joseph Convent High School - 87.14%